Zain Dana Harper

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Curriculum Vitae | Security Evaluation, AI Evaluation, Graphics, and Developer Tools

Profile

Self-directed software builder who writes tools that test AI systems and the software around them, and graphics work that people install. Elder ENB, a lighting preset for Skyrim Special Edition, has 996,649 downloads on Nexus Mods. Flywheel, an AI workstation published on PyPI, runs release checks scored against planted faults. About 13 years of personal programming. Career transition began June 2026 after eleven years of full-time production arboriculture; no professional programming experience.

Seeking red-team, purple-team and security-evaluation work, AI evaluation roles, and graphics engineering. Willing to relocate to London; available for regular travel to San Francisco.

Selected Evidence

- Elder ENB: 996,649 total downloads, 176,359 unique downloads and 2,413 endorsements on Nexus Mods, read October 23, 2026. First uploaded April 21, 2024; last updated February 20, 2026, at version final-elder-9. nexusmods.com/skyrimspecialedition/mods/117327
- Flywheel 1.9.0 behavior-check: flagged 20 of 20 planted behavior changes and 0 of 50 unchanged pairs, on a corpus written before the check ran. Scripted model, Windows only. github.com/HarperZ9/flywheel/releases/tag/v1.9.0
- Daylight Commons slur sign: precision 91.0%, 3,995 of 4,392, with a 95% interval of 90.1 to 91.8 on public human-labelled corpora. github.com/HarperZ9/daylight-commons/blob/main/project-docs/EVALUATION.md
- raw-native: GPU frames matched the CPU reference in 16 of 16 D3D12 and 16 of 16 WebGPU certificates on an RTX 4090. github.com/HarperZ9/raw-native
- Articulate 0.14.0: Markdown masks removed 5,123 findings on table separator rows across 3,000 files and added none. github.com/HarperZ9/articulate/releases/tag/v0.14.0
- Open source: 37 merged engineering pull requests in 30 external repositories, with 17 closed without merge and 14 open, counted from the GitHub API on October 23, 2026. harperz9.github.io/career/open-source-census.json

Self-Directed Projects

About 13 years of self-directed programming | Career transition began June 2026

Sole proprietor since September 2026; no paid software clients

- Own the architecture, implementation, integration, tests, documentation, and release work for Flywheel and related tools. Work spans Python services and command-line tools, Flutter desktop software, Rust systems tooling, and browser interfaces.
- Develop model-agnostic workflows, evaluation harnesses, controlled-action interfaces, and source-linked records that can be checked without relying on a model to approve its own answer.

Personal programming began with game modding, reverse engineering, bot and rotation logic, and runtime-memory tooling, then developed into systems engineering, graphics, compilers, and AI infrastructure.

Freelance Technical Writer, Documentation, and Product Operations

Independent practice | Began 2017; project-based work

- Produced technical and compliance documentation, API and implementation guides, proposals, release notes, and operational playbooks. Turned implementation details into material usable by customers, developers, and reviewers.

Employment

Commercial Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Combined ground-based tree-care work and safety coordination with client intake, site assessments, estimates, proposals, scheduling, and crew/vendor communication.

- Monitored rigging systems, clearances, and changing conditions while supporting climbers from the ground. Explained work options and constraints to clients and maintained continuity between the proposed scope and field execution.

Technical Networking Support | Xbox/Microsoft contract

Subcontracted through Stream/Convergys | Wilsonville, Oregon | 2014 to 2015

- Diagnosed connectivity and account issues by phone and chat using TCP/IP, DNS, NAT, firewall, and router troubleshooting; documented repeatable resolutions and customer-facing instructions.

Security Evaluation: Red Team and Purple Team

No paid red-team or purple-team engagement to date. This work runs on my own systems and owned test targets. Faults and attacks are planted on purpose, so each defensive check is measured against a known answer and its misses stay on the record.

Planted controls in release checks

Released a Flywheel monitor-admission check over 2,676 frozen items with six planted controls, a second independent math kernel, and signer-head anchoring. Release evidence reports no real monitor tested through the check and no installed signer; Windows anchoring was parse-checked only. Behavior-check, below, was scored the same way on planted changes.

Accountable Surface | Governed actions and computer-use adapters

Built operator-granted action interfaces for filesystem, command, API, browser, and Windows UI Automation work. Actions carry explicit scope and postconditions; persisted journals preserve decisions and observed effects. Reversible adapters include restoration paths. Fake drivers make browser and desktop scenarios testable without uncontrolled live actions.

Version 0.2.0 ships source and wheel artifacts on GitHub. The project remains alpha, and a tool refusal or unknown result is never promoted into a successful action.

Phantom | Hardware-identity inspection and reversible system changes

Developed a Rust-based Windows and Linux utility with deterministic profiles, read-only audit, backup-before-write behavior, validation, restoration, and platform packages. The released v1.1.1 scope is userland-reversible identifiers. Kernel and firmware layers are modeled, not shipped.

Controlled assessment and detection engineering

Private systems work includes Array for approval-gated assessment orchestration and containment records, Lab/Seed for controlled-lab fixtures and detection-engineering workflows, and Runtime/ORCA for local workspaces, reports, bundles, and release provenance. Public descriptions cover architecture and safe interfaces; private implementations, target material, and operational details are not published here.

AI Evaluation

Flywheel | Model-agnostic AI workstation and coding harness

Primary owned platform. Flywheel 1.9.0, released October 15, 2026, is on PyPI as flywheel-verify, installs through Homebrew or Scoop, and ships a Windows installer with a bundled engine.

- Routes tasks to hosted or local models through an OpenAI-compatible gateway, with command-line, desktop, and API entry points.
- Provides Rowan, a desktop assistant that translates requests into recorded tasks, together with permission-checked coding and tool-use workflows, session state, and inspectable outcomes.
- Produces sealed, re-derivable receipts and offline MATCH, DRIFT, or UNVERIFIABLE outcomes. A reproducible receipt establishes the bounded check it records, not universal model accuracy or safety.
- Added behavior-check in v1.9.0: it replays 13 scripted sessions on two builds and names the first event that differs. Its planted-change result is listed under Selected Evidence; no hosted model was called.

Terminal State Fixtures and Crucible

Built 324 exhaustive terminal-state fixtures and five synthetic component-evaluation families with 337 cases and 5,058 frozen expectations. Defined controls separate failed outputs, provider failures, and missing evidence from valid results. Executed a pinned METR count_odds interoperability check with three deterministic controls, preserving logs and score provenance through the Inspect-to-Flywheel handoff.

Daylight Commons | Independent, offline-first evidence for readers

Built a browser extension that shows the sourced evidence behind replies on nine platforms: X, YouTube, Reddit, Facebook, Instagram, Bluesky, Threads, TikTok and Mastodon. Each sign names its source, states what it does not prove, and shows its measured precision or "not measured yet"; it never calls an account a bot. No server, account or analytics, no data sold or shared, and saved evidence chains verify offline. Version 3.1.0, released October 23, 2026 under AGPL-3.0-only, ships reproducible packages with signed checksums and a full-source git bundle. No Mozilla-signed build exists yet, and the campaign lens is not yet measured.

Articulate | Writing analysis and accountable editing

Built deterministic prose analysis with register and genre profiles and optional model-assisted editing, through command-line, library, MCP, language-server, and SARIF surfaces. Content-free audit receipts record checks without retaining the prose. Version 0.14.0 is on PyPI as articulate-writing. Version 0.15.0, on GitHub October 23, 2026, keeps the original paragraph when a model rewrite reorders a date or adds a flagged device, and prints the reason.

Graphics and Rendering

Elder ENB | Lighting and post-processing preset for Skyrim Special Edition

Built and maintained an ENBSeries preset with a layered design: weather presets set the atmosphere, and stackable overlays adjust style, lighting, effects and post-processing over a fixed baseline. It ships 38 atmosphere presets for the KreatE weather system, with updates from April 2024 to February 2026. ENBSeries and the required mods belong to their authors. nexusmods.com/skyrimspecialedition/mods/117327

raw-native | C++ renderer with a CPU reference and checked GPU backends

raw-native draws each scene on a CPU reference and on the GPU, then writes a certificate stating whether the two agree within committed tolerances. Release 0.8.0, October 14, 2026, renders through D3D12 and WebGPU and passed 60 of 60 tests with MSVC and GCC; no hardware-GPU render was checked for that release.

Merged since 0.8.0 and not yet released: a Vulkan 1.3 backend whose 600 presented frames on lavapipe in CI match the offscreen and D3D12 WARP renders hash for hash; the painterly shader ported to the C++ core, equal to the JavaScript version across fifteen cases; a tilt-shift lens that repaints out-of-focus areas with coarser brushwork and no blur, about 0.86 ms a frame at 1080p on an RTX 4090; and a character asset toolchain that turns brush strokes into signed distance fields, game meshes, UV atlases and baked maps.

Low Water | Original narrative game in development

An original narrative game written and built on my own engine. All five eras of the story are written, and scripted runs reach the ending on every playable class. Fights run on a deterministic combat simulation whose traces are checked byte for byte. Character art is moving to the first-party asset tools above, with no third-party modelling package. A C++ port onto raw-native is in progress. The repository is private, and no release date or store listing is claimed.

Earlier graphics and color work

Earlier HLSL and C++ modding work covered post-processing, Direct3D 11 integration, shader debugging, GPU traces, and shared-memory interfaces, with public projects SkyrimBridge, ENB Runtime Core, Brender Archival, and Engine Revival. Existing engines and collaborators retain their own attribution.

Color tools include Build Color, Calibrate Pro and Studio Engine: perceptual color spaces, color difference, HDR and tone mapping, and display characterization. No professional calibration certification is claimed.

Developer Tools and Languages

BuildLang | Rust compiler and developer tooling

Developed a systems-language toolchain with typed capability effects, type inference, parser and intermediate-representation work, native C output, shader output, and language-server support. Version 1.8.0, published to crates.io on October 23, 2026, adds Box<T> and recursive types. For 1.7.0, 27 held-out tasks were sealed by hash first; two fresh agents scored 20 and 17 of 27, short of the 22 set in advance, and each failure class became an issue.

EMET | Byte-integrity witnessing

Developed Python, Rust, Node, and Go implementations around shared conformance cases and portable verification records. The witness checks byte-level integrity, anchors, and source correspondence; it does not grant permission, certify semantic truth, or replace an operator decision.

Research intake and workspace tooling

Built Gather for source-carrying research intake, Index for workspace and symbol maps, and Forum for model orchestration with recorded decisions. These tools make source selection, handoffs, and task context inspectable across a multi-repository product. On October 23, 2026, Gather 2.4.0, Crucible 1.6.0, and EMET 1.3.1 shipped to PyPI and Canon 0.7.0 to GitHub. Companion tools include Relay for coding agents, Plexus for tool interfaces, Mneme for source-checked memory, Canon for assistant instructions, and Bulletin, a signed-identity message board; customer-scale deployment is not established.

Selected Open-Source Contributions

Six selected contributions merged across six external repositories, covering bug fixes, tests, and documentation, from 37 merged engineering pull requests in 30 repositories. These are contributions, not employment; the census also records open pull requests and those closed without merge.

- DeepEval #2822: fixed unhashable tool-output and nested-parameter failures in tool-correctness evaluation; merged July 2, 2026. github.com/confident-ai/deepeval/pull/2822
- TOMLKit #549: preserved dotted-key siblings when replacing a value with an array of tables; merged July 14, 2026. github.com/python-poetry/tomlkit/pull/549
- Datasette #2815: corrected malformed composite-key requests from HTTP 500 or incorrect HTTP 200 to HTTP 400; tested too few and too many key components in HTML and JSON. Merged July 7, 2026. github.com/simonw/datasette/pull/2815
- Free Law Project #820: documented runnable database-free fast tests; merged August 25, 2026. github.com/freelawproject/litigant-portal/pull/820
- Hebbian Robotics #157: tested reserved-column drift against the actual queryable episode schema; merged August 25, 2026. github.com/Hebbian-Robotics/hflow/pull/157
- Voxwire #44: added offline regression coverage for a six-event degraded voice turn, matching saved PCM audio and JSONL records to emitted events. Merged August 15, 2026. github.com/vectorvoyager358/voxwire/pull/44

Technical Competencies

Languages and runtimes: Python, Rust, C++, Dart/Flutter, TypeScript/JavaScript, Lua, HLSL, C#, PowerShell, and Bash. Additional cross-language conformance work includes Go and Node implementations.

Security evaluation: planted-fault and control design, approval-gated assessment workflows, detection engineering fixtures, vulnerability management and policy audit, and evidence handling for findings.

AI infrastructure and quality: multi-provider/local-model routing, tool-use loops, task decomposition, MCP, evaluation design, deterministic fixtures, regression testing, worker/verifier separation, failure analysis, and terminal-state scoring.

Verification and controlled actions: capability effects, operator grants, action envelopes, source provenance, hash-chained journals, offline replay, scoped postconditions, conformance suites, and release evidence.

Software delivery: Git/GitHub, GitHub Actions, Python packaging, command-line design, Windows/Linux installers, CMake, test automation, static websites, release documentation, and checksums.

Graphics and domain knowledge: D3D11/D3D12, Vulkan, HLSL, WebGPU/WGSL, ENBSeries presets, painterly and stylised post-processing, signed distance fields, CPU reference rendering and GPU parity checks, color-space and tone-mapping work, GPU instrumentation, and applied arboriculture/horticulture from field practice.

Working Practice

AI-assisted development is part of the workflow. Architecture, integration, source review, test design, release decisions, and responsibility for the submitted work remain mine. Public examples separate observed results from interpretation and retain negative or inconclusive findings.

Research Agenda

My independent research asks what evidence another person or system can check without inheriting an AI system's authority: evaluation provenance, verifier independence, live-state drift, proof-carrying actions, and human authorization for consequential actions. Robotics is a research interest.

AI Accountability, Disclosure, and Public-Interest Writing

Independent essays on AI accountability, disclosure, conflicts of interest, and human consequences, relevant to public-interest and whistleblower-protection work.

- Who Knew First, September 2026. Directed and published an AI-assisted investigation comparing disclosure timelines, evidence access, institutional incentives, and evaluator conflicts across nine reported AI-agent incidents. Separates source claims, inference, and unresolved attribution. harperz9.github.io/who-knew-first.html
- An Open Letter on Checking the Machines, September 2026. Proposed contestable AI decisions, repeatable evidence checks, and accountable responses to objections. harperz9.github.io/checking-the-machines.html
- Borrowed Ground, September 2026. Philosophical essay connecting human interdependence to responsibility, privacy, participation, and the limits of technical evidence as authority over people. Substantial AI assistance disclosed. harperz9.github.io/borrowed-ground.html
- The Sandbox Was Never Just a Box, September 2026. Compared public security reports to examine shared-service exposure and the independence of logs, monitors, and scorers in AI-agent evaluation. harperz9.github.io/the-sandbox-was-never-just-a-box.html

Selected Publications and Research Records

Independent systems papers, preprints, research notes, philosophy papers, a working paper, and archived manuscripts with DOI records. These records are not peer reviewed. An archived philosophical manuscript is not an awarded academic degree.

- EMET: An Authority-Incapable Byte-Level Integrity Witness. Systems paper, 2026. Byte-integrity witnessing and the separation of observation from authority. doi.org/10.5281/zenodo.21230267
- BuildLang: Accountable Compute via Typed Capability Effects. Systems paper, 2026. Capability effects and accountable compilation. doi.org/10.5281/zenodo.21231253
- Witnessed Independence: Recording Whether a Verifier Graded Its Own Work. Preprint, 2026. doi.org/10.5281/zenodo.21232206
- Proof Packets: A Derive-Don't-Trust Envelope for Accountable Agent Actions. Preprint, 2026. doi.org/10.5281/zenodo.21231406
- The Personhood-Gate Handoff. Research note, 2026. doi.org/10.5281/zenodo.21234475
- Re-Perceived Effects. Research note, 2026. doi.org/10.5281/zenodo.21231311
- Faithfulness to a Named Criterion Is the Conserved Quantity Across Lossy Substrates. Research note, 2026. doi.org/10.5281/zenodo.22768398
- The Witnessing Spine. Archived research corpus, 2026. doi.org/10.5281/zenodo.20778927
- Conferred Existence. Long-form philosophical manuscript and archived corpus, 2026. doi.org/10.5281/zenodo.20773724
- Sourcehood Without a Source: The Arity Gap and the Free Will Debate. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126117
- The Seam That Holds While It Is Crossed. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126357
- Self-Given Is Not Self-Grounding. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126458
- The Conservation of Faithfulness. Working paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126484

Education

High School Diploma | Wilsonville High School, Wilsonville, Oregon | 2013

Self-directed study in systems programming, compilers, graphics, color science, AI evaluation, epistemics, and philosophy of mind. Practical arboriculture and plant-science knowledge comes from field work; no separate professional license is claimed.

ORCID: orcid.org/0009-0001-7175-5393 | harperz9.github.io/publications.html | harperz9.github.io/portfolio.html

Credentials and Continuing Education

2026 completions. Verification: harperz9.github.io/career/source-ledger.md.

Qualys Certified Specialist: Policy Audit; Custom Assessment and Remediation; Vulnerability Management Detection and Response; Qualys Kubernetes and Container Security; Qualys TotalCloud Exam; Endpoint Detection and Response | Issued 2026-09-30; expire 2028-09-29

Microsoft Applied Skills | Earned September 2026

- Secure AI solutions in the cloud
- Get started with identities and access using Microsoft Entra
- Secure storage for Azure Files and Azure Blob Storage
- Implement security through a pipeline using Azure DevOps | Intermediate; 95% assessment

Advanced Microsoft Learn Coursework

SC-100: Design security solutions for applications and data. Four module assessments and the case study.

Design solutions that align with MCRA and MCSB. Advanced Microsoft Learn module completed September 30, 2026: security architecture, Zero Trust, insider and supply-chain risk, and AI security controls.

Linux Foundation Course-Completion Certificates

LFEL1012: Secure AI/ML-Driven Software Development; LFS120: Conversational AI: Ensuring Compliance and Mitigating Risks; LFD121: Developing Secure Software

LFS182: Securing Your Software Supply Chain with Sigstore; LFEL1001: Understanding the EU Cyber Resilience Act (CRA); LFS118: Ethical Principles for Conversational AI

LFEL1006: Securing Projects with OpenSSF Scorecard; LFEL1005: Security Self-Assessments for Open Source Projects; LFEL1007: Automating Supply Chain Security: SBOMs and Signatures

LFD125: Security for Software Development Managers; LFS180: Introduction to DevSecOps for Managers; SKF100: OWASP Top 10 Security Threats

LFS184: Introduction to JavaScript Security; LFC108: Cybersecurity Essentials; LFEL1004: Authentication and Authorization for Web/API

LFEL1008: Migrating AI Workflows Across Platforms; LFS147: Introduction to AI/ML Toolkits with Kubeflow

Additional Coursework and Skill Badges

Cisco Networking Academy: Ethical Hacker | Course certificate and learning badge, 2026-09-30; 100% knowledge exam.

Google skill badges, completed 2026-09-30: Google DeepMind: Train A Small Language Model (Advanced; 100/100, four coding checkpoints); Use Agent Skills with Multi-Agent Systems (Intermediate course/Advanced challenge; 100/100, six checkpoints). Multi-agent lab: five Cloud Run services deployed; model-access policy blocked local and cloud generation.

Google BigQuery skill badges, issued 2026-10-01: Perform Predictive Data Analysis in BigQuery (100/100, seven checkpoints; Advanced challenge); Create ML Models with BigQuery ML (100/100, four checkpoints; Intermediate challenge).

Google course-completion badges, completed 2026-09-30: Machine Learning Operations (MLOps) with Agent Platform: Model Evaluation (Intermediate); Google Security Operations - Deep Dive (Advanced; 96% knowledge assessment).

Anthropic Claude Academy completion badges: Model Context Protocol: Advanced topics; Claude Code in action; Building with the Claude API; Claude with Amazon Bedrock | Issued September 30, 2026

UNESCO: Global MOOC on the Ethics of AI | Intermediate course certificate, 2026-09-30; 100%.

BlueDot Impact: AGI Strategy | Course completed in 2026

Developed an AGI Strategy action plan for label-independent AI incident notice, independently reviewable evidence, accountable escalation, and tests for misleading verification results.

Coursework: MCRA/MCSB, Zero Trust, OWASP web Top 10, CRA. Research references: NIST AI RMF/GenAI Profile, ISO/IEC 42001/23894, ISO 37002. Scope: harperz9.github.io/career/standards-reference.md.